Drone-Based Object Detection and Explanation Using Vision-Language Models

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Autonomous drones are increasingly used in surveillance, inspection, search-and-rescue, and environmental monitoring, where reliable object detection is essential. Conventional detectors such as YOLO provide bounding boxes, class labels, and confidence scores, but they do not explain why a detection was made. This paper investigates whether Vision-Language Models (VLMs) can support drone-based object detection by providing both object localization and human-readable explanations. We compare a fine-tuned YOLO11s detector with Qwen3-VL-2B-Instruct and Gemma 4 E2B IT on the VisDrone dataset under two evaluation settings: a *hybrid* setting in which YOLO localizes objects and the VLM explains each detection, and a *pure-VLM* setting in which the VLM performs both localization and explanation without any detector assistance. Detection performance is measured using standard COCO metrics (mAP and recall), while explanation quality is assessed using a five-criterion rubric covering visual grounding, correctness, relevance, absence of hallucination, and usefulness. Our results show that fine-tuned YOLO achieves mAP@0.5 of 0.298, substantially outperforming pure-VLM approaches (Gemma 4: 0.011; Qwen3-VL: 0.023). In the hybrid setting the VLMs produce contextually meaningful explanations, scoring 8.6/10 (Qwen3-VL) and 7.7/10 (Gemma 4) on a five-criterion rubric, yet both fail to localize small, dense, or occluded objects in the pure-VLM setting, confirming that current-generation VLMs are better suited as an explanation layer on top of a conventional detector than as standalone detection systems.

Additional Key Words and Phrases: Drone Vision, Object Detection, YOLO, Vision-Language Models, Explainable AI, VisDrone

1 INTRODUCTION

Unmanned Aerial Vehicles (UAVs) are increasingly deployed in tasks where perception is critical, including infrastructure inspection, traffic monitoring, surveillance, and search-and-rescue operations. Object detection is a central component of such systems, as drones must reliably identify pedestrians, vehicles, cyclists, and other scene elements in real time.

Traditional object detectors, including YOLO-style models, are efficient and accurate for predefined object classes [17]. Their output is typically limited to a bounding box, a class label, and a confidence score. These outputs indicate *what* was detected, *where* it was detected, and *how confident* the model is, but they do not explain *which visual evidence* supported the prediction. In safety-critical or operator-assisted systems, this lack of explanation increases cognitive load, since operators must interpret and verify detections without additional context. This can reduce trust, slow decision-making, and complicate post-hoc failure analysis.

Recent Vision-Language Models (VLMs) can connect visual information with natural language and may support open-vocabulary recognition and explanation. UAV-related work such as Cognitive-Drone demonstrates that vision-language reasoning can improve cognitive drone tasks [14], while LLVM-Drone shows that language models can support UAV visual tasks when combined with reliable lightweight vision models [6]. However, VLMs are substantially more computationally demanding than conventional detectors: even compact models require several seconds per inference call on a dedicated GPU, making real-time deployment on resource-constrained drone hardware impractical with current technology. It also remains unclear how well VLMs can localize objects in aerial imagery and whether their explanations are actually grounded in the image rather than being hallucinated.

This paper investigates the use of VLMs for explainable drone-based object detection. The aim is not only to measure detection accuracy, but also to evaluate whether the explanations produced by VLMs are meaningful and useful for human operators. Specifically, we address three research questions:

- **RQ1:** How accurately can Qwen3-VL-2B-Instruct and Gemma 4 E2B IT localize and classify objects in VisDrone images compared to a YOLO-style baseline, measured using IoU, recall, and mAP?
- **RQ2:** To what extent are the natural-language explanations generated by the selected VLMs correct, visually grounded, and useful for interpreting object detections in drone imagery?
- **RQ3:** Under which aerial image conditions, such as small objects, occlusion, clutter, low resolution, or unusual viewpoints, do VLM-based localization and explanation outputs become unreliable?

The remainder of this paper is organised as follows. Section 2 reviews related work and positions this study against it. Section 3 introduces the detection and vision-language background and defines the evaluation metrics. Section 4 describes the methodology and Section 5 the experimental setup. Section 6 reports the results, Section 7 interprets them, and Section 8 concludes.

2 RELATED WORK

2.1 Drone Imagery and Object Detection

Drone imagery is challenging because objects are often small, occluded, densely distributed, or viewed from unusual angles. Datasets such as VisDrone provide aerial images with bounding box annotations and are widely used to benchmark UAV object detection methods [21]. Object detection is broadly split into two-stage detectors such as Faster R-CNN, which first generate region proposals and then classify them [18], and single-stage detectors such as YOLO, which predict boxes and classes in one pass and are therefore better suited to real-time drone perception [17]; both are limited to fixed class lists and do not explain their predictions. The recurring VisDrone detection challenges document how small, densely

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packed, and occluded objects remain the dominant difficulty in this domain [2], which has motivated multi-scale feature architectures such as Feature Pyramid Networks [10] and is surveyed more broadly for small-object detection [15].

2.2 Vision-Language Models

Vision-Language Models offer a different approach by linking image content to natural language. CLIP introduced strong zero-shot image-text understanding [16], while models such as Grounding DINO extend this idea toward open-vocabulary object detection [13]. SAM further supports region-level analysis through promptable segmentation [8]. These models enable systems to detect objects outside fixed class lists and generate natural-language descriptions of visual content. Instruction-tuned models such as LLaVA extended these capabilities to free-form multimodal dialogue [12], and models such as Qwen2-VL added explicit visual grounding through text-based coordinates [20]; however, such models are prone to object hallucination, describing objects that are not present, a problem first quantified for image captioning [19] and more recently for large VLMs [9], which motivates careful evaluation of explanation faithfulness. Their generality comes at a cost, however: VLMs are large transformer-based models with billions of parameters, requiring significant GPU memory and incurring inference latencies that are orders of magnitude higher than those of specialized detectors such as YOLO.

2.3 VLMs for UAV Applications

Recent UAV research has started to integrate language reasoning into drone systems. CognitiveDrone proposes a Vision-Language-Action model that uses image and language input to generate drone control commands. Its improved version adds a separate reasoning module, improving performance on reasoning, human recognition, and symbol understanding tasks [14]. LLVM-Drone combines large language models with lightweight vision models including YOLO, SAM, RAM, and Grounded SAM. The authors emphasize that large multimodal models can hallucinate and that specialized vision models remain important for reliable perception [6].

Existing UAV-related VLM research primarily focuses on high-level reasoning or combines language models with separate vision modules, rather than directly evaluating whether VLMs can both localize objects and explain their predictions in aerial imagery. This gap motivates the present work. Table 1 positions this study against representative prior work.

Table 1. Positioning of this work against representative prior work. **Aerial** = evaluated on aerial/UAV imagery; **VLM** = uses a vision-language model; **Loc.** = evaluates object localization quantitatively; **Expl.** = evaluates natural-language explanation quality; **Hybrid** = combines a conventional detector with a VLM.

Study	Aerial	VLM	Loc.	Expl.	Hybrid
YOLO [17]	–	–	✓	–	–
CLIP [16]	–	✓	–	–	–
Grounding DINO [13]	–	✓	✓	–	–
CognitiveDrone [14]	✓	✓	–	–	–
LLVM-Drone [6]	✓	✓	✓	–	✓
VisDrone-DET2021 [2]	✓	–	✓	–	–
Small-object survey [15]	✓	–	✓	–	–
LLaVA [12]	–	✓	–	–	–
POPE [9]	–	✓	–	✓	–
This work	✓	✓	✓	✓	✓

3 BACKGROUND

3.1 Object Detection and YOLO

Single-stage detectors predict object locations and classes in one forward pass, trading some accuracy for the speed required by real-time applications. The YOLO family popularised this paradigm by framing detection as direct bounding-box and class regression over an image grid [17]. This study uses YOLO11, a recent Ultralytics iteration with an anchor-free detection head, as its baseline detector [7]. Such detectors are fast and accurate on their trained classes but are restricted to a fixed label set and emit no explanation of their predictions. Aerial imagery is particularly demanding: objects are frequently very small, densely packed, and heavily occluded, which the VisDrone detection challenges identify as the dominant source of difficulty [2] and which recent small-object detection surveys analyse in depth [15].

3.2 Vision-Language Models

Vision-Language Models couple a visual encoder with a language model through a projection module and are driven by natural-language prompts. Instruction tuning on image–text pairs produced general-purpose multimodal assistants such as LLaVA that follow free-form visual instructions [12]. More recent models additionally emit *grounded* outputs: Qwen2-VL and Qwen3-VL are trained to return object coordinates as text and to process images at native, variable resolution [1, 20], while the Gemma line provides compact open multimodal models intended for resource-constrained hardware [4, 5]. Because these models produce coordinates through language modelling rather than a dedicated detection head, their localization behaviour differs fundamentally from YOLO, a contrast this study examines directly. A well-documented failure mode is object hallucination, in which a VLM describes objects that are not present in the image [9].

3.3 Detection Evaluation Metrics

Detection quality is measured with the standard COCO metrics [11], all built on the Intersection-over-Union (IoU) between a predicted

box B_p and a ground-truth box B_{gt}:

$$\mathrm{IoU}(B_p, B_{gt}) = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}. \quad (1)$$

A prediction is a true positive when its IoU with a same-class ground-truth box exceeds a threshold τ, following the matching criterion introduced by the PASCAL VOC challenge [3]. Precision is the fraction of predictions that are correct and recall the fraction of ground-truth objects retrieved:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}. \quad (2)$$

Average Precision (AP) is the area under the precision–recall curve for a class, and mean Average Precision (mAP) averages AP over all classes. We report mAP@0.5 ($\tau = 0.5$), mAP@0.75 ($\tau = 0.75$), and mAP@0.5:0.95 (averaged over $\tau \in \{0.50, 0.55, \ldots, 0.95\}$) following the COCO protocol, together with per-class AP and the size-stratified scores $\mathrm{mAP}_{\mathrm{small}}$, $\mathrm{mAP}_{\mathrm{medium}}$, and $\mathrm{mAP}_{\mathrm{large}}$. All metrics are computed with `pycocotools` after converting predictions and annotations to COCO format.

4 METHODOLOGY

4.1 Dataset

We use the VisDrone-DET dataset [21], which contains real aerial drone imagery annotated with bounding boxes across ten object classes: *pedestrian*, *people*, *bicycle*, *car*, *van*, *truck*, *tricycle*, *awning-tricycle*, *bus*, and *motor*. All experiments use the official validation split of 548 images. Annotations are converted to COCO format for compatibility with standard evaluation tooling.

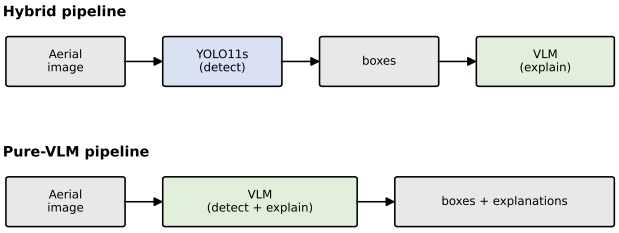


Fig. 1. The two evaluation settings. In the hybrid setting YOLO11s detects objects and the VLM explains each box; in the pure-VLM setting the VLM both detects and explains.

4.2 YOLO Baseline

We evaluate two YOLO configurations as baselines:

- **YOLO11n pretrained on COCO**: a lightweight model with 80 generic classes. Class names are mapped to VisDrone equivalents where possible (e.g., *person* → *pedestrian*).
- **YOLO11s fine-tuned on VisDrone**: YOLO11s fine-tuned on the VisDrone training split; this serves as the primary YOLO baseline. The training procedure is detailed in Section 4.3.

Detection performance is evaluated using pycocotools with standard COCO metrics: mAP@0.5, mAP@0.5:0.95, mAP@0.75, and recall, as well as per-class AP.

4.3 Fine-Tuning the YOLO Baseline

The primary detector is obtained by fine-tuning YOLO11s on VisDrone rather than training from scratch. We start from the COCO-pretrained `yolo11s.pt` checkpoint and continue training on the VisDrone 2019-DET training split, using a YOLO-format version of the dataset whose labels cover the same ten object classes used throughout this study (the VisDrone *ignored-region* and *others* categories are excluded). Transfer learning from COCO lets the model reuse general low-level visual features and adapt only to the aerial domain and the VisDrone label set.

Training uses the Ultralytics framework on a single Kaggle T4 GPU with the following configuration: 50 epochs at an input resolution of 640×640, batch size 16, and a cosine learning-rate schedule. Early stopping is applied with a patience of 15 epochs, so training halts if validation performance does not improve for 15 consecutive epochs. Mosaic augmentation is enabled during training and disabled for the final 10 epochs, allowing the model to converge on undistorted images; all remaining hyperparameters (optimizer, base learning rate, and the other augmentations) are left at the Ultralytics defaults. The checkpoint with the best validation score is retained as the final model.

This fine-tuned model serves both as the standalone detection baseline (RQ1) and as the localizer in the hybrid pipeline, where it supplies the boxes that the VLMs explain. On the 548-image validation split it reaches an mAP@0.5 of 0.298 (Table 2), far above the COCO-pretrained YOLO11n and the pure-VLM models, establishing the upper reference point against which VLM localization is measured.

4.4 VLM Models

The VLM-based part of the study uses two models evaluated in inference-only mode without fine-tuning. Lightweight 2-billion parameter variants were chosen deliberately: real drone hardware has strict size, weight, and power constraints that make deploying large models impractical, so evaluating compact models reflects the conditions under which such a system would realistically operate. In half precision (fp16) both models fit comfortably within the memory budget of a Kaggle T4 GPU (15 GB VRAM); each occupies roughly 5 GB. This is the compute environment used for all experiments; no quantization was required.

- **Qwen3-VL-2B-Instruct**: selected for its explicit support of spatial visual understanding including object positions, viewpoints, occlusions, and grounding [1, 20]. It uses a dynamic resolution patching system that preserves fine-grained detail in high-resolution aerial images.
- **Gemma 4 E2B IT**: the `google/gemma-4-E2B-it` checkpoint, a compact open multimodal model intended for efficient on-device deployment [5], which makes it a realistic lightweight counterpart to Qwen3-VL under the same hardware constraints.

VLM inference is nevertheless far slower than a conventional detector: in the pure-VLM setting a single full-image detection call averages about 38 seconds, so processing the full 548-image validation set requires roughly six hours of continuous GPU time per model. This stands in sharp contrast to YOLO11s, which processes the entire validation set in under one minute. The pure-VLM setting is evaluated on the full 548-image validation split. The hybrid setting, by contrast, is evaluated on a curated 30-image sample, because each generated explanation must be scored by hand against the RQ2 rubric (Section 4.6), which does not scale to the full validation set.

4.5 Evaluation Settings

Two inference settings are evaluated:

Hybrid setting. The fine-tuned YOLO11s first runs on each image and produces bounding box detections. For each detection the VLM receives the original image with the detected bounding box highlighted in red and is prompted to generate a single natural-language explanation describing the visual evidence that supports the detection. This setting isolates explanation quality from localization accuracy.

Pure-VLM setting. The VLM receives the raw aerial image without any detector assistance and is asked to identify all visible objects from the VisDrone class list, outputting bounding box coordinates along with a brief explanation for each detection. This setting evaluates whether VLMs can serve as standalone detection-and-explanation systems.

4.6 RQ2 Evaluation Sample

Explanation quality (RQ2) is evaluated on a curated 30-image sample drawn from the VisDrone validation set. Images are stratified into five scene types to ensure coverage of diverse conditions: core mixed-difficulty, easy single-class, hard dense/occluded, high-resolution, and sparse scenes. These RQ2 scene types are a separate stratification from the RQ3 difficulty groups (Section 6), which are labelled A–E; the two groupings should not be conflated. This sample is used for both VLMs in the hybrid setting.

4.7 Explanation Quality Rubric

To address RQ2, a sample of VLM-generated explanations is manually scored using a five-criterion rubric, each criterion scored 0–2 for a maximum of 10 points per explanation:

(1) **Visual grounding**: does the explanation reference specific visible features such as shape, shadow, colour, or position?
(2) **Correctness**: is the description accurate for the detected object?
(3) **Relevance**: is the explanation focused on the detected object rather than background or other objects?
(4) **No hallucination**: does the explanation avoid invented details not visible in the image?
(5) **Usefulness**: would the explanation help a human operator understand or verify the detection?

4.8 Justification of Design Choices

Each component was chosen to reflect a realistic, resource-constrained drone deployment. **YOLO11s** is the primary baseline because it is a current, lightweight single-stage detector that runs in real time on modest hardware, and fine-tuning it on VisDrone provides a strong domain-specific reference. **Qwen3-VL-2B** and **Gemma 4 E2B** were selected as the two VLMs because both are compact, openly available, instruction-tuned multimodal models that fit a single T4 GPU, with complementary design emphases (Qwen3-VL on explicit spatial grounding, Gemma 4 on efficient on-device multimodality), making them a meaningful contrast under the same constraints. **VisDrone** is used because it is a standard aerial-detection benchmark and exposes the small, dense, and occluded objects central to RQ3. The **hybrid and pure-VLM split** isolates the two questions: the pure setting tests whether a VLM can localise (RQ1), while the hybrid setting removes localisation from the VLM so that explanation quality (RQ2) can be judged independently of detection accuracy. The **30-image RQ2 sample** and **25-image RQ3 set** are hand-stratified across difficulty conditions because both require manual, per-detection judgement that does not scale to all 548 images. Finally, a **manual rubric** rather than an automated text-overlap metric such as BLEU is used for RQ2 because there is no reference explanation to match against, and the qualities of interest (visual grounding and hallucination) cannot be captured by n-gram overlap.

4.9 Statistical Analysis

Detection metrics (RQ1) are reported as point estimates computed with `pycocotools` over the full 548-image validation set. For the explanation rubric (RQ2), each criterion is summarised by its mean score across the $n = 90$ scored explanations per model (the top three detections per image over the 30-image sample). The two models are compared per criterion, and their per-explanation totals are compared with a paired Wilcoxon signed-rank test over the detections scored for both models, reported together with the matched-pairs rank-biserial correlation as an effect size.

5 EXPERIMENTAL SETUP

5.1 Compute Environment

All experiments run in a Kaggle notebook with a single NVIDIA Tesla T4 GPU (15 GB VRAM). YOLO11s is fine-tuned for 50 epochs at 640×640 input resolution. Both VLMs run in half precision (fp16) without quantization; each occupies roughly 5 GB, well within the T4 budget.

5.2 Models and Inference Configuration

The VLMs are the pinned checkpoints `Qwen/Qwen3-VL-2B-Instruct` and `google/gemma-4-E2B-it`, loaded through the Hugging Face `transformers` library with the SDPA attention backend. Decoding is greedy (`do_sample = False`) in every setting. In the hybrid setting the highlighted image is down-scaled to a long side of 1280 px and the explanation is capped at 64 new tokens. In the pure-VLM setting the image is capped at 1280×1280 px and generation at up to 1536 new tokens (detection) or 2048 (detection plus explanation). The exact prompts are reproduced in Appendix A.

5.3 Pipeline and Recorded Outputs

In the hybrid pipeline, YOLO11s detects objects (confidence threshold 0.40) and the VLM then explains each retained box in turn. In the

pure-VLM pipeline, the VLM is queried once per image and its JSON output is parsed into detections. Each model is decoded with a *separate* parser matched to its own coordinate convention (Qwen3-VL `[x1,y1,x2,y2]`, Gemma 4 the Gemini-style `[y1,x1,y2,x2]`; Appendix A); coordinates are returned in a 0–1000 normalised space and rescaled to pixels. We validated this decoding per model: across all 548 images, every one of the 19,587 Qwen3-VL and 7,423 Gemma 4 predicted boxes satisfies $x_1 < x_2$ and $y_1 < y_2$ and lies within the image bounds (zero violations of either condition). This confirms that no coordinate-order transposition occurs and that the low pure-VLM mAP reflects genuine under-detection and loose localisation rather than a parsing artifact. For every image the pipeline records the detections (class, box, confidence), the explanation text, and the per-call latency, and checkpoints results to JSON so that long runs can be resumed.

5.4 Reproducibility

All experiments were run in a Kaggle notebook with a single NVIDIA Tesla T4 GPU. The VLMs are the pinned Hugging Face checkpoints `Qwen/Qwen3-VL-2B-Instruct` and `google/gemma-4-E2B-it`, loaded with the `transformers` library in half precision and without quantization; YOLO11s was fine-tuned with the Ultralytics library. The five pipeline notebooks, the evaluation scripts, and the generated prediction and explanation files are available in the project repository.

6 RESULTS

6.1 Detection Performance (RQ1)

Table 2 shows overall detection metrics across all evaluated models on the VisDrone validation set (548 images). The fine-tuned YOLO11s achieves the highest mAP@0.5 of 0.298, substantially outperforming all other configurations. The COCO-pretrained YOLO11n achieves mAP@0.5 of 0.053, limited by class mapping mismatches and the domain gap between COCO and aerial imagery. Both VLMs in the pure-VLM setting achieve very low mAP@0.5 (Gemma 4: 0.011; Qwen3-VL: 0.023), reflecting fundamental difficulties in producing well-calibrated bounding boxes from language-model inference alone; Qwen3-VL roughly doubles Gemma 4's detection accuracy but both remain an order of magnitude below the fine-tuned YOLO11s.

Table 2. Overall detection metrics on VisDrone validation set (548 images). R = Recall@0.5. Precision@0.5 is omitted: under the COCO definition it is the area under the precision–recall curve, which equals AP@0.5 (the mAP@.5 column), so it carries no additional information. The hybrid setting is omitted: detection is performed by YOLO11s, so its metrics are identical to the YOLO11s row.

Model	mAP@.5	mAP@.5:.95	mAP@.75	R
YOLO11n	0.053	0.036	0.039	0.067
YOLO11s-tuned	0.298	0.188	0.198	0.357
Gemma 4 E2B	0.011	0.004	0.002	0.030
Qwen3-VL-2B	0.023	0.015	0.015	0.056

Across object size bands (Table 3), both YOLO configurations drop sharply on small objects, consistent with the well-known difficulty of small-object detection in aerial imagery, but the VLMs collapse almost entirely: Gemma 4 scores effectively zero and Qwen3-VL only 0.005 $\text{mAP}_{\text{small}}$, indicating that without a detection head small objects are rarely localised at all (Figure 3).

Table 3. Size-stratified detection mAP@0.5:0.95 on the VisDrone validation set (548 images).

Model	$\text{mAP}_{\text{small}}$	$\text{mAP}_{\text{medium}}$	$\text{mAP}_{\text{large}}$
YOLO11n	0.007	0.054	0.242
YOLO11s-tuned	0.094	0.299	0.447
Gemma 4 E2B	0.000	0.003	0.077
Qwen3-VL-2B	0.005	0.031	0.200

Table 4 reports per-class AP at IoU 0.5:0.95. The fine-tuned YOLO11s achieves the highest per-class scores across all categories, with car (0.513) and bus (0.328) scoring highest. VLMs achieve their best relative performance on large, visually distinctive vehicles (car, truck, van) and near-zero scores on small or ambiguous classes (awning-tricycle, tricycle, bicycle).

Table 4. Per-class AP at IoU 0.5:0.95 on the VisDrone validation set. G4 = Gemma 4 E2B pure VLM; Q3 = Qwen3-VL-2B pure VLM; YN = YOLO11n COCO pretrained; YS = YOLO11s VisDrone fine-tuned.

Class	G4	Q3	YN	YS
pedestrian	0.002	0.006	0.032	0.159
people	0.003	0.004	0.000	0.081
bicycle	0.000	0.004	0.009	0.040
car	0.013	0.041	0.185	0.513
van	0.005	0.021	0.000	0.247
truck	0.008	0.040	0.037	0.204
tricycle	0.001	0.002	0.000	0.105
awning-tricycle	0.000	0.000	0.000	0.054
bus	0.005	0.020	0.090	0.328
motor	0.001	0.010	0.009	0.154
mean	**0.004**	**0.015**	**0.036**	**0.188**

6.2 Explanation Quality (RQ2)

We manually scored the top three highest-confidence detections per image across the 30-image RQ2 sample for both models in the hybrid setting (90 explanations per model) using the five-criterion rubric of Section 4 (each criterion 0–2; maximum 10). Table 5 reports the mean score per criterion.

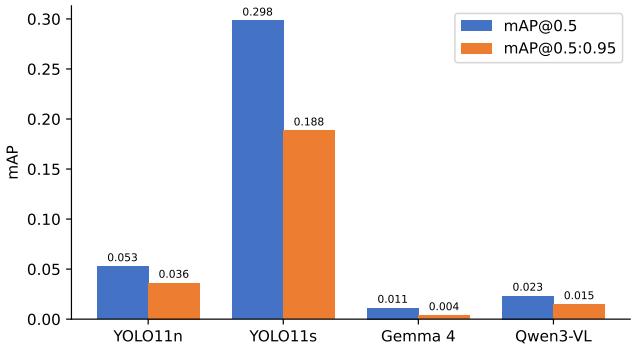


Fig. 2. Detection mAP@0.5 and mAP@0.5:0.95 across the four models (visual summary of Table 2).

Table 5. Mean explanation-quality scores in the hybrid setting (90 explanations per model; each criterion 0–2, maximum 10).

Criterion	Qwen3-VL	Gemma 4
Visual grounding	1.90	1.80
Correctness	1.56	1.46
Relevance	1.79	1.39
No hallucination	1.80	1.76
Usefulness	1.60	1.29
Mean total (/10)	**8.64**	**7.69**

Qwen3-VL scores higher than Gemma 4 on every criterion, with an overall mean of 8.64/10 versus 7.69/10. Both models score highest on visual grounding (1.90 and 1.80), indicating that their explanations typically reference genuinely visible features rather than generic descriptions. Contrary to our initial expectation, no-hallucination is among the *strongest* criteria for both models (1.80 and 1.76), not the weakest: in the hybrid setting the highlighted red box anchors the VLM to a concrete region, which appears to suppress invented detail. The weakest criteria are correctness for Qwen3-VL (1.56) and usefulness (1.29) and relevance (1.39) for Gemma 4, whose explanations more often remain vague or drift to surrounding context. The largest Qwen3-VL advantage is on relevance (1.79 vs. 1.39) and usefulness (1.60 vs. 1.29). Across scene types Qwen3-VL is stable (mean totals 8.3–9.0), whereas Gemma 4 is more variable (6.5–8.5), scoring lowest on the high-resolution group. A paired Wilcoxon signed-rank test over the 86 detections scored for both models confirms that Qwen3-VL's explanations are rated significantly higher than Gemma 4's ($W = 553.5$, $p = 0.001$), with a moderate-to-large effect (matched-pairs rank-biserial correlation $r = 0.47$) and a median advantage of 1.0 point out of 10 per explanation.

6.3 Failure Analysis (RQ3)

We evaluated both pure-VLM models against ground truth on a curated set of 25 VisDrone images (five per group): (A) tiny, dense, low-resolution scenes, (B) occlusion-dominant scenes, (C) extreme object density, (D) mid-difficulty/high-contrast scenes, and (E) sparse scenes (a false-positive probe). For each image we recorded each model's detection count against the ground-truth object count and manually coded the dominant failure mode. Table 6 reports the detection ratios per group.

Table 6. Pure-VLM detections vs. ground truth on the 25-image RQ3 set (mean over 5 images per group). Ratio = detections / ground-truth objects; lower means more severe under-detection. Ratios are computed from unrounded per-group means (e.g. group E's mean ground-truth count is 1.8, displayed as 2).

		Gemma 4		Qwen3-VL	
Group	Condition (mean GT)	ratio	dets	ratio	dets
A	tiny/dense/low-res (111)	4%	4.6	38%	42.4
B	occlusion (45)	21%	9.4	83%	37.2
C	extreme density (114)	17%	19.2	37%	42.6
D	mid-difficulty (33)	26%	8.6	72%	24.0
E	sparse (2)	100%	1.8	89%	1.6

Three findings stand out. First, **both models degrade sharply with object density and small scale**: groups A and C, whose ground-truth counts exceed 110 objects, give the lowest ratios for both models, while the sparse group E (2 objects) is handled almost perfectly (Figures 5 and 6). Second, **Qwen3-VL recovers far more objects than Gemma 4** (group ratios 37–89% vs. 4–26%); Gemma 4 is strongly under-detecting, reporting as few as 4% of the objects present in the densest scenes. Third, this higher recall has a cost: across the 25 images Qwen3-VL's dominant coded failures are *under-detection* (8 images) and *hallucinated objects* (8), whereas Gemma 4's are *imprecise localization* (10) and *under-detection* (8), with only one hallucinated-object case. Gemma 4 is thus conservative (it misses objects and places loose boxes), while Qwen3-VL is aggressive, detecting more but inventing more spurious objects (Figure 7). In the genuinely sparse group E both models were coded *acceptable* on almost every image, indicating that false-positive hallucination is triggered mainly by clutter rather than emitted unconditionally.

Imprecise localization. Even when an object is found, the box is approximated from visual reasoning rather than a learned detection head, producing loose boxes; this is reflected in the collapse from mAP@0.5 to mAP@0.75 for both models (Table 2).

No usable confidence. Neither VLM emits a meaningful confidence score, so the pipeline assigns a constant value of 1.0; confidence thresholding is therefore unavailable as a quality filter. Because every detection shares the same confidence, the precision–recall ranking that average precision integrates over is governed by the model's output order rather than by calibrated certainty, so the reported VLM mAP is partly an artifact of detection ordering.

These failure modes are consistent with the observation in LLVM-Drone [6] that large multimodal models can hallucinate and that specialized vision models remain important for reliable perception in UAV systems.

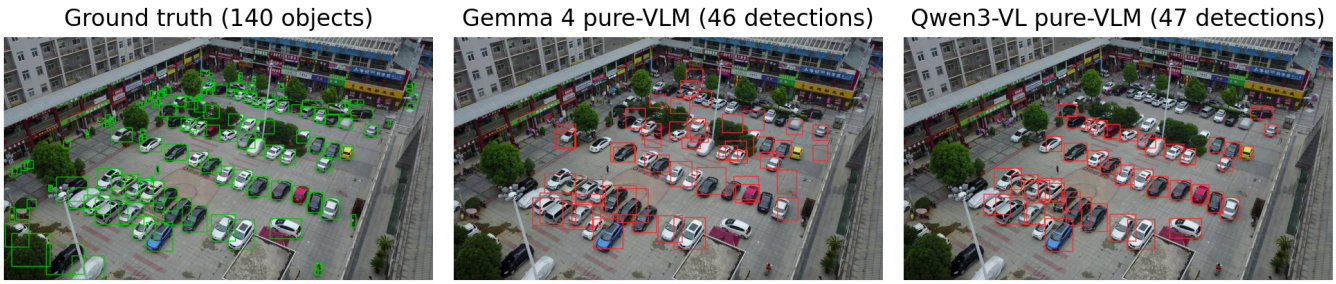


Fig. 3. Under-detection on a dense aerial scene (140 ground-truth objects). Left: ground truth (green). Middle: Gemma 4 pure-VLM (46 detections, red). Right: Qwen3-VL pure-VLM (47 detections, red). Both VLMs recover only about a third of the objects present, illustrating the severe under-detection that drives the low pure-VLM mAP.



Fig. 4. Example explanations. Left: a high-scoring, visually grounded description of a clearly visible car. Right: a low-scoring case where the VLM misidentifies the highlighted object and invents details.

7 DISCUSSION

Across both settings, the results are best understood through a single explanatory account, which we term the *missing detection-head* effect: a VLM approximates object locations through language-level spatial reasoning over a fixed visual-token budget, rather than through a dense, learned detection head. This account predicts the three patterns examined below (severe under-detection in crowded scenes, loose boxes that collapse at higher IoU thresholds, and a steep drop on small objects) and explains why the hybrid setting, which removes localization from the VLM's responsibilities, is the more reliable use of the same model.

7.1 RQ1: Localization Accuracy

The results clearly show that fine-tuned YOLO11s dominates on all detection metrics, with an mAP@0.5 of 0.298 compared to 0.011–0.023 for pure-VLM approaches. This gap is explained by several structural differences. First, YOLO uses a learned detection head that produces calibrated bounding boxes at pixel precision; VLMs approximate positions through language-level spatial reasoning, producing loose boxes. Second, the fine-tuned YOLO benefits from VisDrone-specific training; VLMs receive no domain-specific training in this study. Third, VLMs are not architecturally optimized for exhaustive detection and tend to under-report objects in dense scenes. Fourth, and practically important, VLMs are far slower: a single full-image pure-VLM call averages roughly 38 seconds versus milliseconds for YOLO, ruling out real-time use on current drone hardware entirely.

The COCO-pretrained YOLO11n (mAP@0.5 = 0.053) provides a useful intermediate reference: it outperforms VLMs significantly despite having no VisDrone training, confirming that the VLM gap is not purely a domain-adaptation issue but reflects a fundamental limitation in localization capability.

Qwen3-VL clearly outperforms Gemma 4 on detection (mAP@0.5 0.023 vs. 0.011, and 0.015 vs. 0.004 at mAP@0.5:0.95), roughly doubling its accuracy, though both remain far below the fine-tuned YOLO11s. Both models perform best on large, visually salient vehicles (car, truck, van) and worst on small objects (awning-tricycle, tricycle, bicycle), which is consistent with the hypothesis that VLM localization degrades with object size.

7.2 RQ2: Explanation Quality

The rubric scores confirm that the hybrid setting produces reliable explanations overall: both models average above 7.5/10, and their highest scores are on visual grounding, consistent with the expectation that anchoring the VLM to a YOLO-highlighted region keeps it tied to visible evidence. The expectation that no-hallucination would be the weakest criterion did *not* hold: it is among the strongest for both models, suggesting that hallucination is far less of a problem when the VLM explains a pre-localised box than when it must reason over a full scene (as in the pure-VLM setting). The clearest difference between the models is qualitative: Qwen3-VL's explanations are more relevant and useful (1.79 and 1.60) than Gemma 4's (1.39 and 1.29), which more often describe surrounding context or add little actionable detail. This makes Qwen3-VL the more suitable explanation layer of the two, although both are usable.

7.3 RQ3: Failure Modes

The failure analysis shows that the two VLMs fail in different ways, revealing a recall–precision trade-off. Gemma 4 is conservative: its dominant failures are under-detection and imprecise box placement, so it misses most objects in dense scenes and localises loosely the ones it does find. Qwen3-VL is aggressive: it recovers several times

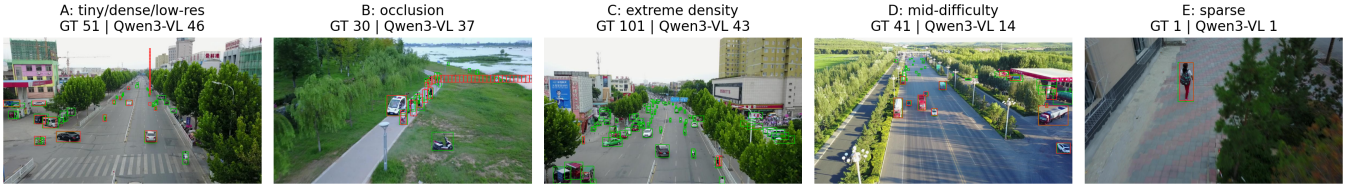


Fig. 5. One representative image per RQ3 difficulty group with ground-truth objects (green) and **Qwen3-VL** pure-VLM detections (red). Under-detection scales with object density (groups A and C); the sparse group E is handled correctly.



Fig. 6. The same five RQ3 scenes as Figure 5 with **Gemma 4** pure-VLM detections (red) against ground truth (green). Gemma 4 under-detects far more severely than Qwen3-VL, recovering only a handful of objects per scene, while likewise handling the sparse group E.

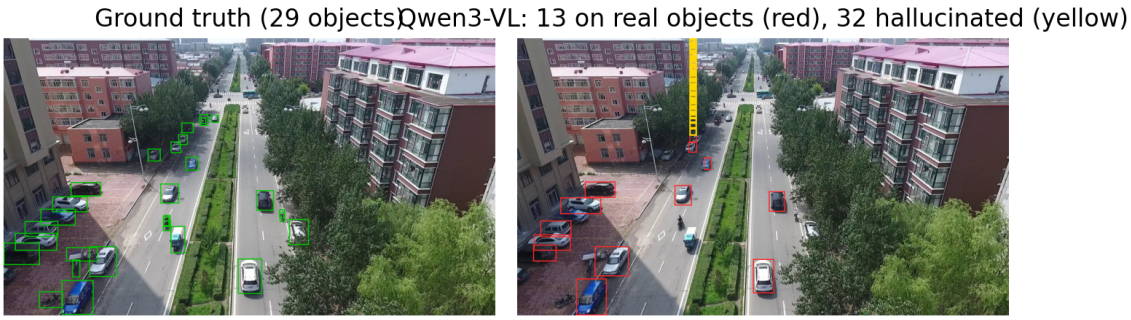


Fig. 7. Hallucination by Qwen3-VL on a mid-difficulty scene (29 ground-truth objects). Left: ground truth (green). Right: Qwen3-VL's 45 detections, split into 13 that overlap a real object (red) and 32 that overlap no ground-truth object at all (yellow): a cluster of invented boxes over an empty strip, while the real vehicles on the road are detected.

more objects, but its second most common failure is hallucinated objects, spurious detections in cluttered scenes. Both models degrade primarily with object density and small scale rather than with occlusion or contrast, and both handle sparse scenes reliably, so density is the dominant predictor of failure. Hallucination is far less prevalent in the hybrid setting (RQ2) because the highlighted box anchors the VLM to a real region; the pure-VLM hallucinations therefore arise from the model having to decide *whether* an object exists, not merely to describe a given one.

These findings reinforce the practical conclusion that VLMs in their current form cannot serve as replacement detectors for aerial surveillance but can serve as an on-demand explanation layer when invoked for specific detections by an operator.

7.4 Practical Implications

The results confirm that VLMs in their current form are unsuitable for real-time or near-real-time drone perception. On a Kaggle T4 GPU, a single pure-VLM detection call averages about 38 seconds for these 2B models at fp16 precision; processing all 548 validation images takes roughly six hours per model. Deploying a VLM on embedded drone hardware, which typically offers a fraction of T4 performance, would increase this latency further and would likely require dedicated edge inference accelerators or model distillation to even approach operational speeds. By contrast, YOLO11s processes an image in under 10 milliseconds on the same hardware, making it orders of magnitude more suitable for real-time deployment.

Despite these computational constraints, VLMs show clear promise as an on-demand explanation layer. In the hybrid setting, a VLM is invoked only when a human operator requests an explanation for a specific YOLO detection, not for every frame in a continuous stream. In this scenario, a latency of several seconds is acceptable, and the computational cost is incurred selectively rather than continuously. This use case is both practical and complementary to

existing detector pipelines, and represents the most realistic near-term deployment model for VLMs in drone surveillance systems.

8 CONCLUSION

This paper quantified how well compact (2B-class) Vision-Language Models localize and explain objects in aerial imagery, comparing Gemma 4 E2B and Qwen3-VL-2B against a fine-tuned YOLO11s baseline on VisDrone under a hybrid setting (YOLO localizes, the VLM explains) and a pure-VLM setting (the VLM does both). The hybrid/pure decomposition is what lets us separate localization ability from explanation quality and attribute each result to the right stage of the pipeline.

The central finding is a sharp asymmetry between these two roles. As standalone detectors the VLMs are not competitive: YOLO11s reaches mAP@0.5 of 0.298 against 0.011–0.023 for the pure-VLM models, and even the COCO-pretrained YOLO11n (0.053), which never saw VisDrone, beats both, so the gap is a fundamental localization limitation, not merely missing domain adaptation, and it is compounded by per-call latency of tens of seconds. As an explanation layer, by contrast, the same models are useful: across 90 hybrid explanations each, Qwen3-VL averages 8.64/10 and Gemma 4 7.69/10, strongest on visual grounding and, contrary to our expectation, low on hallucination once anchored to a YOLO-provided box.

That VLMs belong on top of a detector rather than in place of one is qualitatively consistent with prior work such as LLVM-Drone [6]; the contribution of this study is the quantification and the failure characterisation that the hybrid/pure split makes possible. We show that pure-VLM reliability is governed chiefly by object density and scale (both models collapse on tiny, dense, low-resolution scenes and cope only with sparse ones), and that the two models fail in opposite ways: Gemma 4 is conservative (it under-detects and places loose boxes), whereas Qwen3-VL is aggressive (several times the recall, but hallucinated objects as its second failure mode), a recall–precision trade-off in which neither emits a usable confidence score for filtering. The practical implication is concrete: a hybrid system should keep detection with the specialized model and invoke a VLM on demand to explain a specific detection, where it returns a contextually meaningful, visually grounded description at acceptable, if non-real-time, latency.

8.1 Limitations

Several limitations bound the scope of these findings:

(1) **Model scale.** Only 2-billion-parameter VLMs are evaluated; larger models (such as the 8B variant originally proposed) may localize better, so the results characterise compact, drone-deployable models rather than VLMs in general.
(2) **Single annotator.** The RQ2 rubric scores are assigned by one annotator with no inter-annotator agreement study, so the explanation-quality scores carry unmeasured subjectivity.
(3) **Sample size.** RQ2 uses a 30-image sample and RQ3 a 25-image sample, both stratified by hand; this limits statistical power and the generality of the qualitative findings.
(4) **Prompt design and cross-model prompt differences.** Each setting uses one fixed, hand-written prompt, and the two VLMs are not given an identical prompt: the explanation prompt is more detailed for Gemma 4 (it additionally requests colour, appearance, and activity, and supplies the object's approximate location) than for Qwen3-VL. The detection prompts also use each model's native coordinate convention (Qwen3-VL `[x1,y1,x2,y2]` with a `class` field; Gemma 4 the Gemini-style `[y1,x1,y2,x2]` with a `label` field); this is not a confound, as each output is decoded by a parser matched to its own format and the decoded boxes were validated geometrically (Section 5). Because VLM outputs are otherwise prompt-sensitive, part of the measured difference between the two models may reflect explanation-prompt wording rather than model capability. Prompt variation was otherwise not studied.
(5) **No fine-tuning.** The VLMs are evaluated zero-shot; domain adaptation to aerial imagery was out of scope.
(6) **Uninformative confidence.** Pure-VLM confidence is fixed at 1.0 because the models emit no calibrated score, which prevents confidence thresholding and precision–recall curves.
(7) **Single dataset.** All experiments use VisDrone, so the findings may not transfer to other aerial domains or altitudes.

8.2 Future Work

Future work should explore fine-tuning VLMs on aerial imagery to improve localization accuracy, evaluate explanation quality through a human-subject study to validate the rubric findings, and investigate more efficient model variants (e.g. sub-1B quantized models or specialized aerial VLMs) that could reduce inference latency to a range compatible with operator-in-the-loop workflows.

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A VERBATIM PROMPTS

The prompts below are reproduced verbatim from the experiment notebooks. `{class_name}` is the YOLO class label and `{classes}` the comma-separated list of the ten VisDrone classes; the location placeholders in the Gemma hybrid prompt are filled per detection. Qwen3-VL is prompted for boxes in `[x1,y1,x2,y2]` order, whereas Gemma 4 follows the Gemini-style `[y1,x1,y2,x2]` convention; both use a 0–1000 normalised coordinate space.

A.1 Hybrid: Qwen3-VL

```
This is an aerial drone image. A {class_name}
is highlighted with a red bounding box. In one
sentence, describe the {class_name} you see
inside the red box.
```

A.2 Hybrid: Gemma 4

```
This is an aerial drone image taken from above.
An object detector has identified a {class_name}
and it is highlighted with a red bounding box.
The object is located at approximately {cx}% from
the left and {cy}% from the top of the image,
occupying about {w}% of the image width and {h}%
of the image height.

In one sentence, describe this {class_name} as
you see it -- its colour, what it looks like, and
what it is doing.
```

A.3 Pure-VLM: Qwen3-VL (detection)

```
This is an aerial drone image taken from above.

Detect all visible objects from these
categories: {classes}.

For each object you can see, output a JSON
object with:
  - class: one of the category names above
  - bbox: [x1, y1, x2, y2] normalised to 0-1000
    (0,0 = top-left, 1000,1000 = bottom-right)

Return ONLY a valid JSON array. If no objects
are visible, return [].
Example: [{"class": "car", "bbox": [120,45,200,90]}]
```

In the detection-plus-explanation variant the schema gains an `explanation` field requesting a one-sentence description of each object.

A.4 Pure-VLM: Gemma 4 (detection)

```
This is an aerial drone image taken from above.

Detect all visible objects from these
categories: {classes}.

Return a JSON array where EACH detected object
is a SEPARATE entry with:
  - "label": one of the category names above
  - "box_2d": [y1, x1, y2, x2] normalised to
    0-1000 (top-left = 0,0)

Example format:
[
  {"label": "car",
   "box_2d": [200, 300, 250, 400]},
  {"label": "pedestrian",
   "box_2d": [100, 150, 180, 200]}
]

Do NOT group multiple objects under one entry.
One object per entry. Return ONLY a valid JSON
array. If no objects are visible, return [].
```

The detection-plus-explanation variant adds an `explanation` field to each entry, requesting a one-sentence description.

B DECLARATION ON THE USE OF AI TOOLS

Generative AI tools were used while preparing this thesis. In line with the University of Twente's policy on the use of AI in student work, and as agreed with the supervisor, this section sets out what they contributed. Their role was confined to support and implementation; the ideas, decisions, and claims are the author's own.

In writing the manuscript, the tools reorganised the author's notes into a coherent outline, corrected grammar and phrasing in

passages the author had already drafted, flagged inconsistencies and gaps for the author to resolve, and turned the author's numerical results into the LaTeX tables and the Matplotlib and PIL figures. On the implementation side, they helped scaffold and debug the Python notebooks and evaluation scripts and drafted code comments, always following an experimental design fixed in advance by the author.

The research questions, the experimental design, and the execution of every experiment (the YOLO11s fine-tuning and both the hybrid and pure-VLM pipelines) were the author's own work, as were the manual RQ2 rubric scoring and the RQ3 failure-mode coding, which used no automated assistance. Each piece of AI-generated text, code, and figure was checked against the underlying data and the author's intent before it was used, and the author takes full responsibility for the methodology, results, and conclusions of this thesis.

C DETAILED RESULTS TABLES

These tables give the detailed breakdowns underlying the summary results in Section 6.

Table 7. RQ2 explanation scores: mean ± standard deviation per criterion (90 explanations per model; each criterion 0–2).

Criterion	Qwen3-VL	Gemma 4
Visual grounding	1.90 ± 0.34	1.80 ± 0.54
Correctness	1.56 ± 0.54	1.46 ± 0.62
Relevance	1.79 ± 0.44	1.39 ± 0.61
No hallucination	1.80 ± 0.45	1.76 ± 0.57
Usefulness	1.60 ± 0.51	1.29 ± 0.59

Table 8. RQ2 mean explanation total (/10) by scene type.

Scene type	Qwen3-VL	Gemma 4
core / mixed-difficulty	9.00	7.60
easy / clear	8.27	7.67
hard / small / occluded	8.80	8.33
high-resolution	8.27	6.47
sparse	8.53	8.47